

Kamal Pangen, Ph.D.

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Professional Profile

Applied AI/ML Lead with a PhD in theoretical physics and 8+ years of industry experience building machine learning and GenAI solutions, including agentic and RAG systems, across financial services and healthcare. Trained as a researcher, I go beyond building with LLMs to understand how they work, choose the right approach for each problem, and optimize and validate them properly. I bring a strong quantitative foundation, hands-on Python expertise, and experience translating technical results for executive and non-technical audiences.

Experience

Applied AI/ML Lead @JPMorgan Chase May 2025–Present

- Designed and deployed an end-to-end eDiscovery pipeline combining embedding-based retrieval with LLM prompting to isolate relevant document populations from thousands of files, saving millions in external legal review fees.
- Extended retrieval pipeline to automatically extract key case events and generate structured timeline narratives, eliminating manual legal document review.
- Architected a tool-calling conversational agent that autonomously queries structured databases, joining multi-source data to output precise, natural language responses to user prompts.

Senior Data Scientist @CVS Health Oct 2021–May 2025

- Built an LLM agent to evaluate member prior authorization eligibility against electronic provider records, accelerating authentication precision and turnaround time.
- Developed a LangChain GenAI engine to auto-generate structured test/question sets from policy documents, reducing processing time from days to under an hour.
- Created an anomaly detection algorithm uncovering generic drug pricing exploitation by select pharmacies; partnered with pricing teams to remediate and received the **Premier Award**.
- Engineered a semi-supervised pipeline (t-SNE clustering with XGBoost/SHAP personas) identifying fraudulent pharmacy entities for escalation to special investigation teams.

Data Scientist - Machine Learning @Mercy Hospital, St. Louis Feb 2018–Sept 2021

- Built and deployed a clinical NLP model predicting inpatient DRG billing codes from clinical notes, generating \$10M+ in annual revenue capture; named innovation award finalist.
- Applied NLP algorithms on imaging/pathology reports and physician notes to automate cancer registry case-finding and optimize inpatient diagnosis code workflows.
- Modeled the financial impact of elective surgery cancellations during COVID-19 to advise C-suite leadership on resource allocation and operational strategy.

Skills

Languages: Python, SQL, C++, Java, MATLAB, Mathematica, $\LaTeX$

Data & Analytics: BigQuery, Pandas, Matplotlib, Seaborn, QlikSense, QlikView, Tableau

AI / Machine Learning: PyTorch, TensorFlow, Keras, Scikit-Learn, XGBoost, LightGBM, Vertex AI, OpenAI, RAG, LLM Agents, LangChain

Education

Washington University in St. Louis, St. Louis, MO

Ph.D. in Physics

AUG 2017

M.S. in Physics

MAY 2013

Brigham Young University, Provo, UT

B.S. in Physics and Mathematics

AUG 2011

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Machine Learning Certificates

- **Building Basic Generative Adversarial Networks** JUN 2023
- **Causal Inference from Observational Data** JAN 2022
- **Reinforcement Learning Specialization** (4-course sequence) MAR 2020
- **Introduction to TensorFlow for AI, ML, and DL** JUN 2019
- **Deep Learning Specialization** (5-course sequence) MAY 2019
- **How to Win a Data Science Competition** JUN 2018
- **Machine Learning** NOV 2017

Honors and Awards

- **Premier Award**, CVS Health AUG 2023
- **Finalist**, CVS Analytic Venture Competition (AVC2022) Nov 2022
- **Finalist**, Mercy-Wide Innovation Conference OCT 2019
- **Winner**, 1st Mercy Hackathon [\[Link\]](#) SEPT 2019
- **Arthur L. Hughes Fellow**, Washington University [\[Link\]](#) SUMMER 2012
- **University Fellow**, Washington University [\[Link\]](#) 2011, 2013
- **Harvey Fletcher Scholarship**, Brigham Young University 2010–2011
- **Academic & Physics Department Scholarships**, Brigham Young University 2007–2010

Publications

(Note: In high energy physics, authors are listed in alphabetical order)

- M. Alford, **K. Pangeni**, and A. Windisch, “Color Superconductivity and Charge Neutrality in Yukawa Theory”, *Physical Review Letters* 120, 082701 (2018).
- H. Nishimura, M. Ogilvie, and **K. Pangeni**, “Liquid-Gas Phase Transition and CK Symmetry in Quantum Field Theories”, *Physical Review D* 95, 076003 (2017).
- M. Alford and **K. Pangeni**, “Gap bridging enhancement of modified Urca process in nuclear matter”, *Physical Review C* 95, 015802 (2017).
- H. Nishimura, M. Ogilvie, and **K. Pangeni**, “Complex mass spectrum in lattice QCD with static quarks at strong coupling”, *Physical Review D* 93, 094501 (2016).
- H. Nishimura, M. Ogilvie, and **K. Pangeni**, “Complex saddle points and disorder lines in QCD at finite temperature and density”, *Physical Review D* 91, 054004 (2015).
- H. Nishimura, M. Ogilvie, and **K. Pangeni**, “Complex saddle points in QCD at finite temperature and density”, *Physical Review D* 90, 045039 (2014).